

Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 10 multiple-choice questions (1 mark each). Part B: 5 structured questions (27 marks). Same format as the final exam.

Answer Part A by circling one option. Answer Part B in full, showing your working. A periodic table and the answer key are at the back.

Part A · Multiple choice

Choose one answer for each question.

A1 Topic: Ionic and Covalent Compound II IC2-LO1

A covalent bond is formed when two atoms _____ a pair of electrons.

- A. transfer
- B. share
- C. lose
- D. gain

A2 Topic: Ionic and Covalent Compound II IC2-LO3

The number of lone pairs on the nitrogen atom in NH_3 is _____.

- A. 0
- B. 1
- C. 2
- D. 3

A3 Topic: Ionic and Covalent Compound II IC2-LO2

The correct name for N_2O_5 is _____.

- A. nitrogen pentoxide
- B. dinitrogen pentoxide
- C. dinitrogen tetroxide
- D. nitrogen(V) oxide

A4 Topic: Ionic and Covalent Compound II IC2-LO2

The correct formula for sulfur hexafluoride is _____.

- A. SF_6
- B. S_6F
- C. SF_4
- D. S_2F_6

A5 Topic: Ionic and Covalent Compound II IC2-LO3

The total number of valence electrons in the Lewis structure of HCN is _____.

- A. 8
- B. 10
- C. 12
- D. 14

A6 Topic: Ionic and Covalent Compound II IC2-LO3

A double bond consists of _____ shared electrons.

- A. 2
- B. 4
- C. 6
- D. 8

A7 Topic: Ionic and Covalent Compound II IC2-LO4

Which bond is the most polar? (EN: H 2.1, C 2.5, N 3.0, O 3.5, F 4.0)

- A. C-H
- B. N-H
- C. O-H
- D. F-F

A8 Topic: Ionic and Covalent Compound II IC2-LO5

Which molecule has a trigonal pyramidal shape?

- A. BF_3
- B. NH_3
- C. CH_4
- D. H_2O

A9 Topic: Ionic and Covalent Compound II IC2-LO5

Which molecule is nonpolar?

- A. H_2O
- B. NH_3
- C. CO_2
- D. HCl

A10 Topic: Ionic and Covalent Compound II IC2-LO6

Hydrogen bonding occurs between molecules in which hydrogen is bonded directly to _____.

- A. carbon
- B. chlorine
- C. nitrogen, oxygen or fluorine
- D. sulfur

Part B • Structured questions

Show all working. The marks and Bloom's level are shown for each question.

Question B1 Topic: Ionic and Covalent Compound II IC2-LO2 5 marks, C2

A hospital pharmacy stores several gases and solvents. (a) Name the covalent compounds N_2O (used as an analgesic gas), SO_2 and CCl_4 . (3 marks) (b) Write the molecular formulas of dinitrogen tetroxide and phosphorus pentachloride. (2 marks)

Question B2 Topic: Ionic and Covalent Compound II IC2-LO3 6 marks, C2

Hydrogen cyanide (HCN) is a highly toxic gas, and pharmacists must know it for poisoning cases. Carbon is the central atom. (a) Draw the Lewis structure of HCN, showing how you counted the electrons. (4 marks) (b) State the shape of HCN and whether it is polar. (EN: H 2.1, C 2.5, N 3.0) (2 marks)

Question B3 Topic: Ionic and Covalent Compound II IC2-LO5 5 marks, C3

Chloroform (CHCl_3) was once used as a general anaesthetic, and carbon tetrachloride (CCl_4) was once used as a cleaning solvent. Both have carbon as the central atom. Determine the shape of each molecule and explain why chloroform is polar but carbon tetrachloride is nonpolar. (EN: H 2.1, C 2.5, Cl 3.0)

Question B4 Topic: Ionic and Covalent Compound II IC2-LO6 6 marks, C2

A 3% hydrogen peroxide solution (H_2O_2) is used as a mild antiseptic. The atoms are joined H-O-O-H. (a) Calculate the molecular mass of H_2O_2 (H = 1.01, O = 16.00). (2 marks) (b) Draw the Lewis structure of H_2O_2 . (2 marks) (c) Identify the strongest intermolecular force between H_2O_2 molecules and justify your answer. (2 marks)

Question B5

Topic: Ionic and Covalent Compound II

IC2-LO6

5 marks, C3

Fluorine (F_2), hydrogen chloride (HCl) and hydrogen fluoride (HF) are small molecules of similar size. Identify the strongest intermolecular force in each, then arrange them in order of increasing boiling point with a justification.

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Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01 2.1																	2 He 4.00 —
3 Li 6.94 1.0	4 Be 9.01 1.5											5 B 10.81 2.0	6 C 12.01 2.5	7 N 14.01 3.0	8 O 16.00 3.5	9 F 19.00 4.0	10 Ne 20.18 —
11 Na 22.99 0.9	12 Mg 24.31 1.2											13 Al 26.98 1.5	14 Si 28.09 1.8	15 P 30.97 2.1	16 S 32.07 2.5	17 Cl 35.45 3.0	18 Ar 39.95 —
19 K 39.10 0.8	20 Ca 40.08 1.0	21 Sc 44.96 1.3	22 Ti 47.87 1.5	23 V 50.94 1.6	24 Cr 52.00 1.6	25 Mn 54.94 1.5	26 Fe 55.84 1.8	27 Co 58.93 1.9	28 Ni 58.69 1.9	29 Cu 63.55 1.9	30 Zn 65.38 1.6	31 Ga 69.72 1.6	32 Ge 72.63 1.8	33 As 74.92 2.0	34 Se 78.97 2.4	35 Br 79.90 2.8	36 Kr 83.80 —
37 Rb 85.47 0.8	38 Sr 87.62 1.0	39 Y 88.91 1.2	40 Zr 91.22 1.4	41 Nb 92.91 1.6	42 Mo 95.95 1.8	43 Tc [98] 1.9	44 Ru 101.07 2.2	45 Rh 102.91 2.2	46 Pd 106.42 2.2	47 Ag 107.87 1.9	48 Cd 112.41 1.7	49 In 114.82 1.7	50 Sn 118.71 1.8	51 Sb 121.76 1.9	52 Te 127.60 2.1	53 I 126.90 2.5	54 Xe 131.29 —
55 Cs 132.91 0.7	56 Ba 137.33 0.9	57–71	72 Hf 178.49 1.3	73 Ta 180.95 1.5	74 W 183.84 1.7	75 Re 186.21 1.9	76 Os 190.23 2.2	77 Ir 192.22 2.2	78 Pt 195.08 2.2	79 Au 196.97 2.4	80 Hg 200.59 1.9	81 Tl 204.38 1.8	82 Pb 207.2 1.9	83 Bi 208.98 1.9	84 Po [209] 2.0	85 At [210] 2.2	86 Rn [222] —
87 Fr [223] 0.7	88 Ra [226] 0.9	89–103	104 Rf [267] —	105 Db [268] —	106 Sg [269] —	107 Bh [270] —	108 Hs [277] —	109 Mt [278] —	110 Ds [281] —	111 Rg [282] —	112 Cn [285] —	113 Nh [286] —	114 Fl [289] —	115 Mc [290] —	116 Lv [293] —	117 Ts [294] —	118 Og [294] —
Lanthanides		57 La 138.91 —	58 Ce 140.12 —	59 Pr 140.91 —	60 Nd 144.24 —	61 Pm [145] —	62 Sm 150.36 —	63 Eu 151.96 —	64 Gd 157.25 —	65 Tb 158.93 —	66 Dy 162.50 —	67 Ho 164.93 —	68 Er 167.26 —	69 Tm 168.93 —	70 Yb 173.05 —	71 Lu 174.97 —	
Actinides		89 Ac [227] 1.1	90 Th 232.04 1.3	91 Pa 231.04 1.4	92 U 238.03 1.4	93 Np [237] —	94 Pu [244] —	95 Am [243] —	96 Cm [247] —	97 Bk [247] —	98 Cf [251] —	99 Es [252] —	100 Fm [257] —	101 Md [258] —	102 No [259] —	103 Lr [262] —	

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

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Part A: 10 multiple-choice questions (1 mark each). Part B: 5 structured questions (27 marks). Same format as the final exam.

Answer key

Part A · Multiple choice

A1 Topic: Ionic and Covalent Compound II IC2-LO1

A covalent bond is formed when two atoms _____ a pair of electrons.

- A. transfer
- B. share**
- C. lose
- D. gain

Sharing a pair of electrons is the definition of a covalent bond.

A2 Topic: Ionic and Covalent Compound II IC2-LO3

The number of lone pairs on the nitrogen atom in NH_3 is _____.

- A. 0
- B. 1**
- C. 2
- D. 3

N: 5 valence electrons, 3 used in N-H bonds, 2 left = 1 lone pair.

A3 Topic: Ionic and Covalent Compound II IC2-LO2

The correct name for N_2O_5 is _____.

- A. nitrogen pentoxide
- B. dinitrogen pentoxide**
- C. dinitrogen tetroxide
- D. nitrogen(V) oxide

di- for 2 N, pent(a)- for 5 O; the vowel of penta- is dropped before oxide.

A4 Topic: Ionic and Covalent Compound II IC2-LO2

The correct formula for sulfur hexafluoride is _____.

A. SF₆

B. S₆F

C. SF₄

D. S₂F₆

No prefix on sulfur = 1 S; hexa- = 6 F.

A5 Topic: Ionic and Covalent Compound II IC2-LO3

The total number of valence electrons in the Lewis structure of HCN is _____.

A. 8

B. 10

C. 12

D. 14

H 1 + C 4 + N 5 = 10.

A6 Topic: Ionic and Covalent Compound II IC2-LO3

A double bond consists of _____ shared electrons.

A. 2

B. 4

C. 6

D. 8

Two shared pairs = 4 electrons.

A7 Topic: Ionic and Covalent Compound II IC2-LO4

Which bond is the most polar? (EN: H 2.1, C 2.5, N 3.0, O 3.5, F 4.0)

A. C-H

B. N-H

C. O-H

D. F-F

Differences: 0.4, 0.9, 1.4, 0. O-H is the largest.

A8 Topic: Ionic and Covalent Compound II IC2-LO5

Which molecule has a trigonal pyramidal shape?

- A. BF_3
- B. NH_3**
- C. CH_4
- D. H_2O

3 bonds + 1 lone pair on N.

A9 Topic: Ionic and Covalent Compound II IC2-LO5

Which molecule is nonpolar?

- A. H_2O
- B. NH_3
- C. CO_2**
- D. HCl

CO_2 is linear; its two $\text{C}=\text{O}$ dipoles point in opposite directions and cancel.

A10 Topic: Ionic and Covalent Compound II IC2-LO6

Hydrogen bonding occurs between molecules in which hydrogen is bonded directly to _____.

- A. carbon
- B. chlorine
- C. nitrogen, oxygen or fluorine**
- D. sulfur

Only H-N , H-O and H-F give the strong $\delta+$ hydrogen needed.

Part B · Structured questions

Question B1

Topic: Ionic and Covalent Compound II

IC2-LO2

5 marks, C2

A hospital pharmacy stores several gases and solvents. (a) Name the covalent compounds N_2O (used as an analgesic gas), SO_2 and CCl_4 . (3 marks) (b) Write the molecular formulas of dinitrogen tetroxide and phosphorus pentachloride. (2 marks)

(a) N_2O : two N, one O → dinitrogen monoxide. SO_2 : one S (no mono-), two O → sulfur dioxide. CCl_4 : one C, four Cl → carbon tetrachloride. (b) di- = 2 N, tetr- = 4 O → N_2O_4 . No prefix = 1 P, penta- = 5 Cl → PCl_5 .

Step	Marking criteria	Marks
1	N_2O = dinitrogen monoxide	1
2	SO_2 = sulfur dioxide	1
3	CCl_4 = carbon tetrachloride	1
4	dinitrogen tetroxide = N_2O_4	1
5	phosphorus pentachloride = PCl_5	1
	Total	5

Question B2 Topic: Ionic and Covalent Compound II IC2-LO3 6 marks, C2

Hydrogen cyanide (HCN) is a highly toxic gas, and pharmacists must know it for poisoning cases. Carbon is the central atom. (a) Draw the Lewis structure of HCN, showing how you counted the electrons. (4 marks) (b) State the shape of HCN and whether it is polar. (EN: H 2.1, C 2.5, N 3.0) (2 marks)

Valence electrons: $H\ 1 + C\ 4 + N\ 5 = 10$. Skeleton $H-C-N$ uses 4; the other 6 go on N as three lone pairs. C now has only 4 electrons, so two lone pairs of N become bonding pairs: $H-C\equiv N:$. Check: 10 electrons in total, H has 2, C has 8, N has 8 (6 in the triple bond + 1 lone pair). Shape: C has 2 bonding groups and no lone pairs $\rightarrow$ linear. $H-C$ ($\Delta EN\ 0.4$) and $C\equiv N$ ($\Delta EN\ 0.5$) both pull electron density towards N; the dipoles point the same way and do not cancel, so HCN is **polar**.

Step	Marking criteria	Marks
1	Total valence electrons = $1 + 4 + 5 = 10$	1
2	Skeleton $H-C-N$ with C central	1
3	Triple bond between C and N	1
4	One lone pair on N; H has 2 and C and N have 8 electrons	1
5	Shape: linear (2 groups, no lone pairs on C)	1
6	Polar: dipoles point the same way towards N and do not cancel	1
	Total	6

Question B3

Topic: Ionic and Covalent Compound II

IC2-LO5

5 marks, C3

Chloroform (CHCl_3) was once used as a general anaesthetic, and carbon tetrachloride (CCl_4) was once used as a cleaning solvent. Both have carbon as the central atom. Determine the shape of each molecule and explain why chloroform is polar but carbon tetrachloride is nonpolar. (EN: H 2.1, C 2.5, Cl 3.0)

Both molecules have 4 bonding groups and no lone pairs on C, so both are **tetrahedral** (109.5°). C-Cl bonds are polar ($\Delta\text{EN } 0.5$). In CCl_4 the four identical C-Cl dipoles are arranged symmetrically and cancel $\rightarrow$ nonpolar. In CHCl_3 one position has H instead of Cl; C-H is almost nonpolar ($\Delta\text{EN } 0.4$), so the three C-Cl dipoles are not balanced $\rightarrow$ a net dipole $\rightarrow$ polar.

Step	Marking criteria	Marks
1	Both have 4 bonding groups and 0 lone pairs on C	1
2	Both shapes are tetrahedral	1
3	C-Cl bond is polar ($\Delta\text{EN} = 0.5$)	1
4	CCl_4 : four identical dipoles cancel by symmetry $\rightarrow$ nonpolar	1
5	CHCl_3 : C-H differs from C-Cl, so dipoles do not cancel $\rightarrow$ polar	1
	Total	5

Question B4

Topic: Ionic and Covalent Compound II

IC2-LO6

6 marks, C2

A 3% hydrogen peroxide solution (H_2O_2) is used as a mild antiseptic. The atoms are joined H-O-O-H. (a) Calculate the molecular mass of H_2O_2 ($\text{H} = 1.01$, $\text{O} = 16.00$). (2 marks) (b) Draw the Lewis structure of H_2O_2 . (2 marks) (c) Identify the strongest intermolecular force between H_2O_2 molecules and justify your answer. (2 marks)

(a) $2 \times 1.01 + 2 \times 16.00 = 2.02 + 32.00 = 34.02$ amu. (b) Valence electrons = $2 + 12 = 14$. Three single bonds H-O, O-O, O-H use 6; the remaining 8 go as two lone pairs on each O. Each O has 8 electrons, each H has 2. (c) Hydrogen bonding: each H is bonded directly to a small, very electronegative O atom, and each O has lone pairs to attract the δ^+ H of a neighbouring molecule.

Step	Marking criteria	Marks
1	Correct set-up: $2 \times 1.01 + 2 \times 16.00$	1
2	Molecular mass = 34.02 amu	1
3	14 valence electrons; skeleton H-O-O-H with single bonds	1
4	Two lone pairs on each O (octets; H has 2)	1
5	Strongest force: hydrogen bonding	1
6	Justification: H bonded directly to O, which has lone pairs	1
	Total	6

Question B5

Topic: Ionic and Covalent Compound II

IC2-LO6

5 marks, C3

Fluorine (F_2), hydrogen chloride (HCl) and hydrogen fluoride (HF) are small molecules of similar size. Identify the strongest intermolecular force in each, then arrange them in order of increasing boiling point with a justification.

F_2 is nonpolar (identical atoms): dispersion forces only. HCl is polar (ΔEN 0.9) but H is bonded to Cl: dipole-dipole forces. HF has H bonded directly to F: hydrogen bonding. Stronger attractions need more energy to separate the molecules, so $F_2 < HCl < HF$. (Measured values: F_2 about $-188\text{ }^\circ\text{C}$, HCl about $-85\text{ }^\circ\text{C}$, HF about $20\text{ }^\circ\text{C}$.)

Step	Marking criteria	Marks
1	F_2 : dispersion forces only (nonpolar)	1
2	HCl: dipole-dipole forces	1
3	HF: hydrogen bonding	1
4	Order: $F_2 < HCl < HF$	1
5	Justification: stronger intermolecular force $\rightarrow$ more energy needed $\rightarrow$ higher boiling point	1
	Total	5

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